Public Economics

Lecture 1 · Why Study Public Finance?

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Today's Questions

- What is public finance, and what are the key questions the field addresses?
- What are the key facts about the size and growth of government, and the distribution of taxes and spending, here in Austria?
- How do the questions of public finance show up in a concrete policy debate — the Covid-19 pandemic?

Based on
Gruber, J. *Public Finance and Public Policy*, Ch. 1 — "Why Study Public Finance?"

Outline

A Motivating Case: Covid-19

2020: A Natural Experiment in the Role of Government

- March 16, 2020: Austria is among the first countries in Europe to enter a full nationwide lockdown.
- *Kurzarbeit* (short-time work) is massively expanded to keep workers attached to their employers.
- A series of relief instruments follows: *Fixkostenzuschuss*, *Ausfallsbonus*, the Corona-Hilfsfonds.
- February 2022: Austria becomes the *first EU country* to legislate a general Covid-19 vaccination mandate.

Every one of these decisions is a public finance decision.

Source: Covid-19-Maßnahmengesetz, BGBl. I Nr. 12/2020; SN.at, "16. März 2020: Der Tag, ab dem Österreich stillstand"; Impfpflichtgesetz, BGBl. I Nr. 4/2022.

1.1 The Four Questions of Public Finance

What Is Public Finance?

Public finance
The study of the role of **government** in the economy.

The four questions of public finance:

1. **When** should the government intervene in the economy?
2. **How** might the government intervene?
3. What is the **effect** of those interventions on economic outcomes?
4. **Why** do governments choose to intervene in the way that they do?

Q1: When Should the Government Intervene?

Baseline: in a competitive market, the equilibrium (supply = demand) maximizes the *social efficiency* of trade.

- Every trade valued by both sides happens; no valuable trade is missed.
- So the free market is the benchmark — government intervention needs a *justification*.

Two justifications for intervention

1. **Market failure** — the market outcome is not efficient
2. **Redistribution** — society wants a different *distribution* of the (efficient) outcome

Market Failure: The Health Insurance Example

Market failure
A problem that causes the market economy to deliver an outcome that does not maximize efficiency.

- Health insurance *looks* like a textbook competitive market: many buyers, many sellers.
- But: if I skip a flu shot, I raise *my* risk of getting the flu — and the risk that I infect *you*.
- I only weigh *my own* costs and benefits when deciding whether to insure/vaccinate; I ignore the cost I impose on others.
- The same logic applies to vaccination generally: below some coverage threshold, *herd immunity* breaks down and outbreaks become possible again — private vaccination choices under-provide protection relative to the social optimum.

$\Rightarrow$ a **negative externality**: private decisions $\neq$ socially efficient decisions.

Austria's Case: The 2022 Covid-19 Vaccination Mandate

- Feb 2022: Austria's *Impfpflichtgesetz* made Covid-19 vaccination compulsory for adults — a first in the EU.
- Justified explicitly on **externality** grounds: protecting the health system and others, not just oneself.
- June 2022: the *Verfassungsgerichtshof* (Constitutional Court) upheld the law *in principle*, but applied strict scrutiny because a vaccine mandate is a serious intervention in bodily autonomy.
- The mandate was never enforced with fines and was repealed in 2023 as Omicron changed the risk calculus.

Why this matters for public finance

A textbook market-failure argument (Q1) still had to survive constitutional *and* political scrutiny (Q4) before becoming policy (Q2) — and it was reversed once the underlying facts changed (Q3).

Source: Verfassungsgerichtshof, *Impfpflichtgesetz* decisions G 29/2022, G 37/2022 (2022)

The Second Motive: Redistribution

Redistribution
The shifting of resources from some groups in society to others.

- Think of the economy as a **pie**: efficiency determines its *size*.
- Society may also care about the *distribution* of slices — e.g., valuing €1 more to a poor household than to a rich one.

The equity–efficiency trade-off
Redistribution usually changes behavior (e.g., work incentives), which can *shrink the pie* even as it changes how it is sliced.

Why Redistribution Is on the Table in Austria

Income inequality

- Gini coefficient (equivalized disposable income): 30.7 in 2021, up from 29.8 in 2020
- At-risk-of-poverty rate: $\approx$ 12–14% of the population
- Material deprivation nearly doubled: 201,000 people (2.3%) in 2022 $\rightarrow$ 336,000 (3.7%) in 2023

Wealth inequality

- Median household net wealth: $\approx$ €83,000
- Top 1% hold roughly 40%+ of total net wealth
- Bottom 50% hold under 4% of total net wealth

Source: Statistik Austria, EU-SILC (2023); Oesterreichische Nationalbank, Household Finance and Consumption Survey (HFCS).

Q2: How Might the Government Intervene?

Four broad policy tools:

1. **Tax or subsidize** private sale/purchase (change the price)
2. **Restrict or mandate** private sale/purchase (change the quantity directly)
3. **Public provision** (the government produces the good itself)
4. **Public financing of private provision** (government pays; private firms produce)

The Four Tools, With Austrian Examples

Tool	Austrian example
Tax/subsidize	*Pendlerpauschale* (commuter tax deduction); CO_2 pricing under the eco-social tax reform
Mandate/restrict	Covid-19 vaccination mandate (2022, repealed 2023)
Public provision	ÖGK public health insurance; public hospitals (Gemeindespital network)
Public financing of private provision	Social-insurance reimbursement of private physicians and private hospitals

Q3: What Are the Effects of Interventions?

Direct effects
What would happen *if people did not change their behavior*.

Indirect effects
What happens *because* people change their behavior in response.

Both matter for costing *any* government intervention — here is an Austrian example.

Worked Example: The Austrian *KlimaTicket*

- Launched Oct. 2021: one nationwide public-transport ticket at a much lower price than existing regional/ÖBB passes combined.
- **Direct-cost estimate**: priced in only existing pass holders switching to the cheaper ticket. Net cost, first full year (2022): €192.2 million.
- **Indirect effect**: take-up far exceeded forecasts — 215,000+ tickets sold by Dec. 2022, past 300,000 by 2023 (Upper Austria alone: 3.5× the projection).
- Budgeted cost roughly *tripled* within two years — €366.7m (2023), €544.5m (2024) — as ÖBB long-distance ridership rose 20% above its 2019 level.

How many people actually switch is an empirical question — and it was most of the bill.

Source: profil.at; derStandard.at; ORF; Rechnungshof, Bericht KlimaTicket (2025).

Application: Independent Scorekeepers

Answering Q3 requires credible, nonpartisan empirical analysis — Austria has two dedicated institutions for this.

- **Fiskalrat**: monitors compliance with EU and national fiscal rules.
- Parliament's **Budgetdienst**: independently costs and analyzes budget bills for MPs.
- Both flagged the 2024 deficit overshoot well before the official numbers were finalized.

(Comparable independent bodies exist elsewhere too — e.g. the US Congressional Budget Office.)

Source: Fiskalrat-Büro, Pressemitteilung "Aktueller Budgetausblick 2024 und 2025" (5 Nov. 2024); Parlament Österreich, Budgetdienst, "Aktuelle Themen zur Budgetentwicklung 2024 und 2025."

Q4: Why Do Governments Do What They Do?

- Q2 was **normative** (how *should* government intervene?). Q4 is **positive** (why do governments actually behave as they do?).
- Governments aggregate the preferences of millions of citizens — imperfectly.
- Just as markets can fail, **governments can fail** too: policy can reflect lobbying, electoral incentives, or institutional constraints rather than pure efficiency or equity goals.

Political economy
The theory of how the political process produces decisions that affect individuals and the economy.

Same virus, same evidence — yet Austria's own coalition governments split sharply over how far to go.

Normative vs. Positive Public Economics

Normative
Analysis of how things *should* be.

Should the government have mandated Covid-19 vaccination? *How high* should the top income tax rate be?

Positive
Analysis of how things *really are*.

Did the Impfpflicht actually raise vaccination rates? *Does* Kurzarbeit actually prevent layoffs, or mostly subsidize jobs that would have survived anyway?

Positive analysis (mostly empirical) is the necessary first step before normative analysis (mostly theoretical) can be completed.

1.2 Facts on Government in Austria

Why Look at the Facts?

- Public finance is compelling because government is *everywhere* in economic life.
- We now look at six facts about Austria: size, decentralization, deficits/debt, spending composition, revenue composition, and regulation.
- Where useful, we place Austria in its EU/OECD context — the environment you will actually work and pay taxes in.

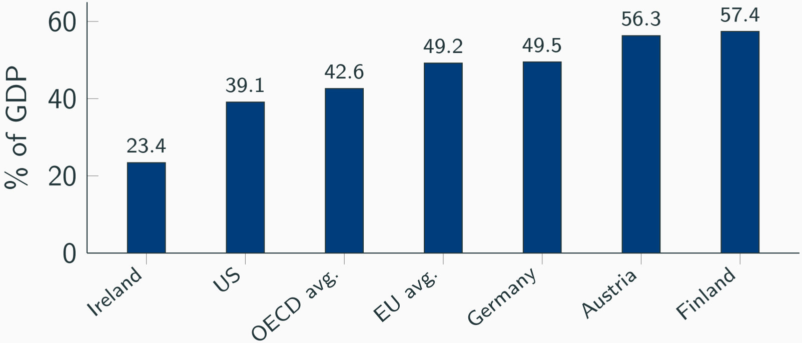
Fact 1: The Size of Government Has Grown — Everywhere

- Government spending has expanded across Western Europe since 1960, driven above all by the build-out of the welfare state.
- The pace has varied enormously: Greece's government share roughly tripled since 1960; Sweden's peaked near two-thirds of GDP in the early 1990s before falling back.
- Austria's own **Staatsquote** sits well above the OECD average today (next slide).

Discussion
What explains the twentieth-century growth of government spending across almost every developed country?

Source: Gruber, *Public Finance and Public Policy*, Fig. 1-2, citing OECD (2021), Table 29.

Government Spending Across the EU — and the US



Austria (**56.3%**) is well above the EU/OECD averages and the US (39.1%) — and close to Finland, one of the EU's highest.

Source: Ireland, Finland: IMF (2024 projections). US, OECD avg.: OECD *Government at a Glance 2025* (2023). EU avg.: Eurostat (2024). Germany: Destatis (2024). Austria: Statistik Austria (2024).

Fact 2: Decentralization — Who Spends, Who Taxes?

Austria, like Germany, is organized as a federal state with three levels: *Bund*, *Länder*, and *Gemeinden*.

Austria's federalism paradox

- Over **90%** of tax revenue is collected centrally as shared federal taxes (*gemeinschaftliche Bundesabgaben*) and redistributed to Länder/Gemeinden as *Ertragsanteile*, via formulas fixed in the *Finanzausgleichsgesetz*.
- Austria's Länder raise only ≈**0.82%** of GDP through their *own* taxes — versus ≈9% of GDP in Germany — down from 4.2% in 1990.
- The Bund transfers ≈**€32.5 billion** a year to Länder and Gemeinden, ≈€10 billion of it to municipalities.

Extensive spending autonomy, almost no revenue autonomy — a classic "who pays vs. who decides" problem in fiscal federalism.

Source: KDZ – Zentrum für Verwaltungsforschung, "Österreichs Föderalismus: OECD-Vergleich zeigt Reformbedarf"; profil.at, "Finanzausgleich: Riesensummen fließen an Länder und Gemeinden."

Why So Centralized? A Political-Economy View

- Redistribution is hard to sustain at the local level: if a *Land* or *Gemeinde* taxes high earners heavily on its own, they can simply move to a lower-tax neighbor.
- Centralizing revenue at the **Bund** is what makes Austria's *Finanzausgleich* possible — transferring resources from richer to poorer regions in a way a fully decentralized system could not.
- This is a Q4 question, not just an accounting fact: **who benefits from more vs. less centralization is a political, not just technical, choice.**

(Same logic globally: parties that prioritize redistribution tend to favor centralization; parties skeptical of redistribution tend to favor devolving power to lower levels.)

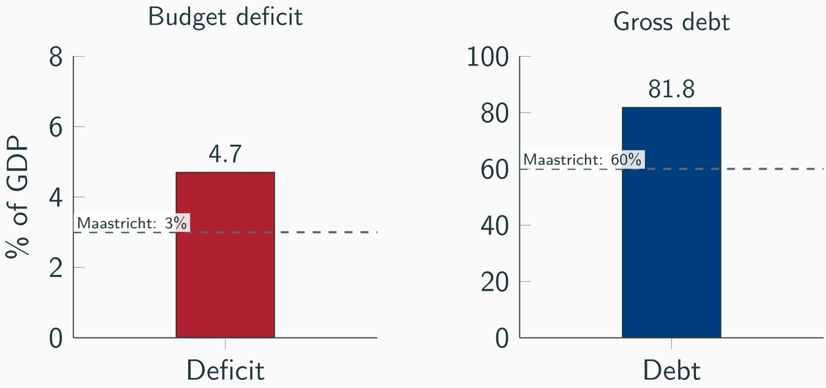
Fact 3: Deficits and Debt

Definitions
Deficit = spending − revenue in a given year. **Debt** = accumulated past deficits.

- The EU's **Maastricht rules** cap the deficit at 3% of GDP and gross debt at 60% of GDP for member states.
- Austria's Länder and Gemeinden are constitutionally close to balanced-budget requirements; the Bund is not.

Source: Protocol (No. 12) on the Excessive Deficit Procedure, Treaty on the Functioning of the EU; Austrian *Bundes-Verfassungsgesetz*, Art. 13 (2).

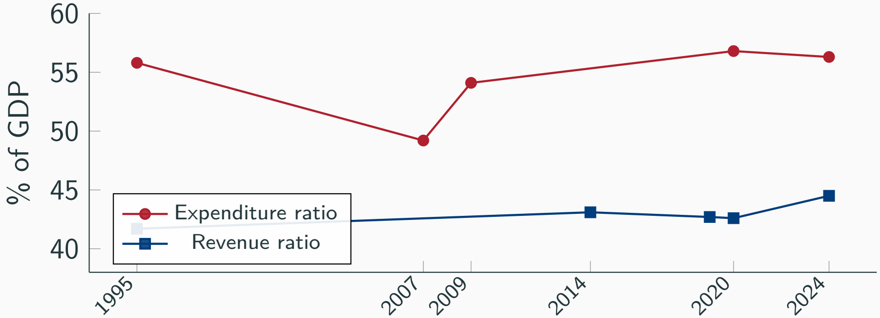
Austria vs. the Maastricht Rules



Austria 2024: deficit **4.7%** of GDP, gross debt **81.8%** of GDP — both above their Maastricht ceilings (3% and 60%).

Source: Statistik Austria, Maastricht notification (2025); BMF, Öffentliche Schulden 2024.

Austria's Revenues and Expenditures Over Time



Expenditure spikes in crises (2009, 2020) and partly recedes; revenue moves in a narrower band — the gap between the lines *is* the deficit.

Source: Fiskalrat, Jahresbericht 2024; Statistik Austria; WKO-Statistik Langzeitreihen; WIFO-Monatsberichte 2015(2).

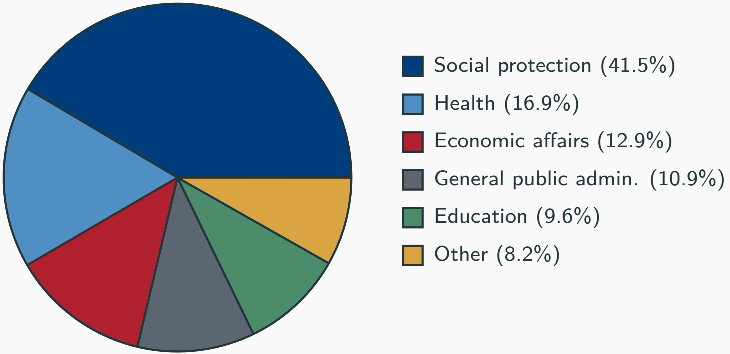
Discussion: Deficits and Debt

- What are the costs of running larger deficits and carrying more debt?
- Why might it make sense to let deficits rise in *some* years (recessions, pandemics) even under fixed fiscal rules?
- Austria's Länder and Gemeinden are far closer to balanced budgets than the Bund — why might that be, given Fact 2?

Fact 4: What Does Government Spend On?

Public goods vs. social insurance
National defense is a classic **public good** (non-excludable, non-rival benefit) — and a small share of Austria's budget. Pensions and health coverage are **social insurance**: government provision of insurance where private insurance markets under-perform — and they dominate Austrian spending.

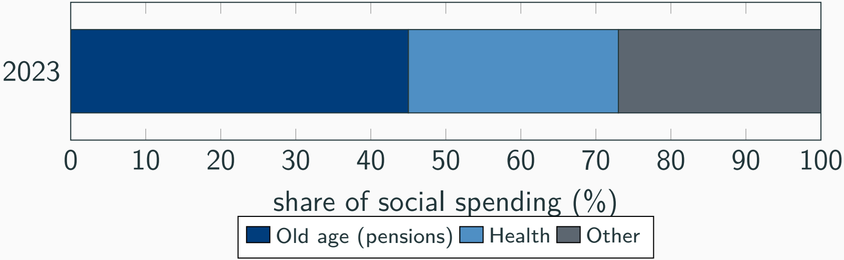
Austria's Government Expenditure, by Function



Total general government expenditure: €271.3 billion (2024), COFOG classification.

Source: Statistik Austria, "Öffentliche Finanzen 2024" (press conference, 31 March 2025).

Austria's Social Spending



"Other" = family benefits, unemployment insurance, disability, and housing support.

Total social spending: €146.2 billion (2023). Austria's **Sozialquote**: 30.9% of GDP — among the highest social-spending ratios in the world.

Source: Statistik Austria, ESSPROS social protection statistics (2023).

Fact 5: Where Does the Revenue Come From?

Across many advanced economies, tax structures have shifted over recent decades — away from taxing corporate profits, toward taxing labor income and consumption.

Discussion
What are the implications of shifting the tax burden from businesses toward workers' earnings?

Austria's Tax Mix: Heavy on Labor

- VAT (*Umsatzsteuer*): ≈€35.4 billion (2022) — the single largest tax
- Wage tax (*Lohnsteuer*): ≈€25 billion — about one-sixth of total tax revenue
- Corporate tax (*Körperschaftsteuer*): ≈€7.8 billion — only 2.2% of GDP
- VAT + wage tax alone ≈ two-thirds of all tax revenue

A distinctive feature

Austria taxes **labor** more heavily, relative to the size of its economy, than any other EU country — while capital and consumption taxes play a comparatively smaller role.

Source: Agenda Austria, "Steuern und Abgaben auf einen Blick"; A&W-Blog, "Steuerlandschaft Österreich: Land der Berge – Land der Täler"; BMF tax revenue statistics.

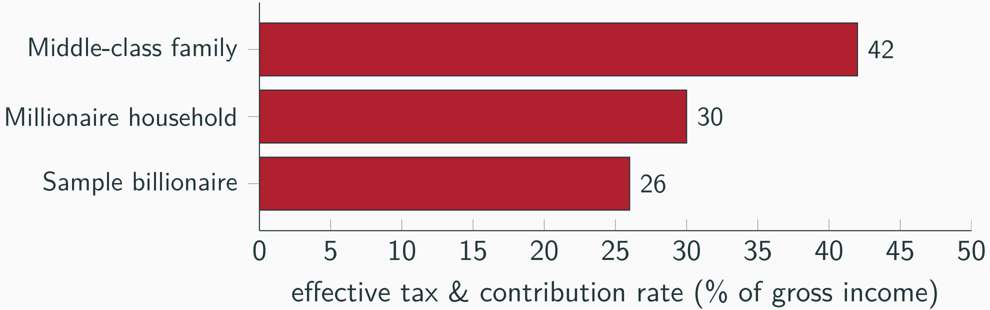
Austria's Own Income-Distribution Data

- Austria has its own "Distributional National Accounts" (DINA) research, in the tradition of Piketty–Saez–Zucman's work for the US.
- Jestl & List (2020) construct such series for Austria, 2004–2016.
- Finding: before the 2008 financial crisis, pre-tax income for the **top 10%** grew faster than for the rest of the population; after the crisis, top-10% income growth turned *negative* while the rest of the distribution stagnated.

Two very different decades for the same income group — growth and crisis do not raise all incomes equally.

Source: Jestl & List, "Distributional National Accounts (DINA) for Austria, 2004–2016," wiiw/WIFO Working Paper 175 (2020); World Inequality Lab Working Paper 2020/23.

Austria's Tax System, Top to Bottom



The statutory top income-tax rate is 55%—but a billionaire's *income* is mostly business profit and capital gains, taxed on different (lower, flatter) terms than wages.

Source: Momentum Institut, "Problem Steuerstruktur: Arbeit hoch besteuert, Vermögen niedrig," Policy Brief 2022.01; "Niedrigsteuerland für Superreiche."

Fact 6: The Regulatory State

Beyond taxing and spending, governments **regulate**.

Domain	**Austrian regulator**
Food & drug safety	AGES (Agentur für Gesundheit und Ernährungssicherheit)
Workplace safety	Arbeitsinspektorat; AUVA
Environment	Umweltbundesamt
Communications & media	RTR (Rundfunk und Telekom Regulierungs-GmbH)
Financial markets	FMA (Finanzmarktaufsicht)

Regulatory agencies can be tiny in budget yet govern a huge share of everyday economic life.

1.3 Back to Covid-19: The Four Questions in Practice

The Four Questions, Applied

1. **When intervene?** Mask mandates, business closures, and the vaccine mandate were all justified by *externalities* — exactly the logic of the flu-shot example.
2. **How intervene?** Kurzarbeit (public financing of private employment) vs. the Fixkostenzuschuss (a direct subsidy) — two different tools for the same goal.
3. **What effect?** Did Kurzarbeit actually prevent layoffs, or mostly subsidize jobs that would have survived anyway? This is exactly a direct-vs.-indirect-effects question.
4. **Why this policy?** Austria's own coalition partners split sharply over vaccine-mandate enforcement — despite drawing on the same epidemiological evidence.

1.4 Conclusion

Highlights

- Government intervention is justified by **market failure** (grow the pie) or **redistribution** (reslice the pie) — not simply by good intentions.
- There are many tools to intervene: taxes/subsidies, mandates/restrictions, public provision, and public financing of private provision.
- Empirical public finance — direct vs. indirect effects — is essential to judging *how well* any policy actually works.
- Governments can fail too: political economy explains why actual policy often departs from the efficiency/equity ideal.
- Government in Austria is large by international standards (56.3% of GDP), fiscally federal in a distinctive way, and its tax base leans heavily on labor income.

Discussion Questions for Next Time

1. Is Austria's heavy reliance on labor taxation (vs. capital/consumption) efficient, equitable, both, or neither?
2. Austria's Länder raise only $\approx$0.82% of GDP through their own taxes, versus $\approx$9% of GDP in Germany. What are the risks of this mismatch between taxing and spending authority?
3. Consider the case for a wealth tax in a country where the top 1% hold $\approx$40% of net wealth. Frame your answer using the equity–efficiency trade-off.
4. Was the 2022 Covid-19 vaccination mandate a case of correcting a market failure, or a case of government overreach? Use the four questions of public finance to structure your answer.
5. If Austria closes its budget deficit mainly through higher consumption taxes rather than income or wealth taxes, low-income households bear roughly $3\times$ the burden (relative to income) of the top decile. Is this an efficiency argument, an equity argument, or both?

Sources (1/3): Textbook, Legislation & Institutions

Textbook: Gruber, J. *Public Finance and Public Policy*, Ch. 1, Worth Publishers (Fig. 1-2, citing OECD (2021), Table 29).

Legislation & official bodies:

- Covid-19-Maßnahmengesetz, BGBl. I Nr. 12/2020
- Impfpflichtgesetz, BGBl. I Nr. 4/2022; Verfassungsgerichtshof, decisions G 29/2022, G 37/2022 (2022)
- Protocol (No. 12) on the Excessive Deficit Procedure, Treaty on the Functioning of the EU; Austrian *Bundes-Verfassungsgesetz*, Art. 13(2)
- Fiskalrat-Büro, Pressemitteilung "Aktueller Budgetausblick 2024 und 2025" (5 Nov. 2024); Fiskalrat, Jahresbericht über die öffentlichen Finanzen 2023–2028 (Dec. 2024)
- Parlament Österreich, Budgetdienst: "Aktuelle Themen zur Budgetentwicklung 2024 und 2025"

Sources (2/3): Statistics

- Statistik Austria: Staatsausgaben nach Aufgabenbereichen (COFOG); “Öffentliche Finanzen 2024” press conference (31 Mar. 2025); Sozialquote/ESSPROS (2023); EU-SILC income and living conditions (2023); Maastricht notification (2025)
- Bundesministerium für Finanzen (BMF): Öffentliche Schulden 2024; tax revenue statistics (2022–24)
- Oesterreichische Nationalbank (OeNB): Household Finance and Consumption Survey (HFCS)
- Agenda Austria: “Steuern und Abgaben auf einen Blick” (tax structure)
- KDZ – Zentrum für Verwaltungsforschung: “Österreichs Föderalismus: OECD-Vergleich zeigt Reformbedarf”; profil.at: “Finanzausgleich: Riesensummen fließen an Länder und Gemeinden”
- A&W-Blog: “Steuerlandschaft Österreich: Land der Berge – Land der Täler”
- WKO-Statistik, Langzeitreihen (Abgabenquote); WIFO-Monatsberichte 2015(2), “Abgabensystem: Status quo”
- Eurostat, government finance statistics (2024); OECD *Government at a Glance 2025* (2023 data); IMF, World Economic Outlook (2024 projections); Statistisches Bundesamt (Destatis, 2025)

Sources (3/3): Media & Distributional Research

- Rechnungshof, Bericht KlimaTicket (2025); profil.at; derStandard.at; ORF (KlimaTicket coverage)
- SN.at, “16. März 2020: Der Tag, ab dem Österreich stillstand”
- Momentum Institut: “Problem Steuerstruktur: Arbeit hoch besteuert, Vermögen niedrig” (Policy Brief 2022.01); “Niedrigsteuerland für Superreiche”; “Budgetsanierung durch Konsumsteuern ist zu Lasten Einkommensärmerer”
- Jestl & List, “DINA for Austria, 2004–2016,” wiiw/WIFO Working Paper 175 (2020)

Two slides in this lecture (Austria’s income/tax-distribution data) are modeled on Emmanuel Saez’s Econ 131 (UC Berkeley) introductory lecture, itself following Piketty–Saez–Zucman’s World Inequality Database methodology.

Thank you!

Questions and discussion